

Empirical Benchmarking Apparatus

Interpretability Score (S) as a Quantitative Metric - standard vs. adversarial training; 7 dataset-model pairings x 2 regimes x 2 runs = 28 runs.

Metrics tracked throughout: Accuracy, PGD Accuracy (robustness), S / Structural Bottleneck Score (interpretability), Integral, Generalization (accuracy-PGD gap)

1. **Summary table (mean +/- CI) - accuracy, PGD accuracy, S, integral, per pairing, standard vs. adversarial**
2. **Best vs. last checkpoint table - same metric set, stability/drift over training**
3. **Training curves - small multiples: rows = pairings, columns = metrics, standard vs. adversarial with variance bands**
4. **Stability-across-runs plot - individual-run traces plus bold mean, per metric, per pairing**
5. **Compute table - wall-clock time per run, per pairing/regime**
6. **Ablation - on the key hyperparameter for computing S (layer/rank/PCA-truncation choice), reporting effect on all four metrics**
7. **Significance test - paired test (Wilcoxon/paired t) per pairing, per metric**
8. **Trade-offs section - 8a Accuracy vs. S; 8b PGD accuracy vs. S; 8c Accuracy vs. PGD accuracy; 8d 3-way scatter matrix**
9. **Full Observation Visualization - every summary number backed by its raw observations:**
 - 9.1 Per-pairing distribution plot (violin/strip) - one panel per metric, raw points overlaid
 - 9.2 Eigenvalue spectrum overlay - the raw covariance spectrum S is derived from, per pairing
 - 9.3 Training-curve panel grid - observation-level version of (3)
 - 9.4 Stability strip - observation-level version of (4)
 - 9.5 Relationship scatter matrix - accuracy x robustness x interpretability, all runs plotted
 - 9.6 Best-vs-last paired dot plot - drift shown as a slope, per metric, per pairing
 - 9.7 Layer-wise score matrix (heatmap) - standard, adversarial, and difference map
 - 9.8 Layer x width x integral relationship - heatmap plus line-slice view by depth bin
 - 9.9 Depth/width marginal plots - score/integral vs. depth and vs. width
 - 9.10 Layer-to-network aggregation check - whole-network S vs. mean/min/max/last-layer aggregations
 - 9.11 Layer importance / contribution ranking - which layers drive the aggregate score
 - 9.12 Cross-layer correlation matrix - modularity vs. shared structure across layers
 - 9.13 Layer-wise training trajectory - one panel per layer, S vs. epoch
 - 9.14 Bottleneck localization plot - where the rank-loss/concentration bottleneck sits
 - 9.15 Per-layer robustness-interpretability scatter - layer-level version of (8b)
10. **Leaderboard-style cross-model comparison - answers "which model wins" directly, the way RobustBench leaderboards do:**
 - 10.1 Ranked dot plot - all pairings/regimes sorted by S, mean with 95% CI
 - 10.2 Multi-metric parallel coordinates - one line per pairing/regime across accuracy, PGD accuracy, S, integral
 - 10.3 Pareto frontier plot - robustness vs. interpretability, Pareto-optimal pairings highlighted
 - 10.4 Rank-stability plot - leaderboard order, best vs. last checkpoint

Model / Dataset / Method Pairings

Model	Dataset	Regime
ResNet-18	CIFAR-10	Standard
ResNet-18	CIFAR-10	Adversarial
VGG-16	CIFAR-10	Standard
VGG-16	CIFAR-10	Adversarial
ResNet-18	CIFAR-100	Standard
ResNet-18	CIFAR-100	Adversarial
WideResNet-28-10	CIFAR-100	Standard
WideResNet-28-10	CIFAR-100	Adversarial
ResNet-18	TinyImageNet-200	Standard
ResNet-18	TinyImageNet-200	Adversarial
ResNet-18	CUB-200-2011	Standard
ResNet-18	CUB-200-2011	Adversarial
ViT-B/16	CUB-200-2011	Standard
ViT-B/16	CUB-200-2011	Adversarial

7 pairings, ResNet-18 as the through-line across all four datasets, plus one dataset-specific specialist each (VGG-16, WideResNet-28-10, ViT-B/16). 7 pairings x 2 regimes x 2 replicate runs = 28 runs.

Table 1

Pairing	Regime	Accuracy (%)	PGD Accuracy (%)	S	Integral
ResNet-18/CIFAR-10	Standard	— ± —	— ± —	— ± —	— ± —
ResNet-18/CIFAR-10	Adversarial	— ± —	— ± —	— ± —	— ± —
VGG-16/CIFAR-10	Standard	— ± —	— ± —	— ± —	— ± —
VGG-16/CIFAR-10	Adversarial	— ± —	— ± —	— ± —	— ± —
ResNet-18/CIFAR-100	Standard	— ± —	— ± —	— ± —	— ± —
ResNet-18/CIFAR-100	Adversarial	— ± —	— ± —	— ± —	— ± —
WideResNet-28-10/CIFAR-100	Standard	— ± —	— ± —	— ± —	— ± —
WideResNet-28-10/CIFAR-100	Adversarial	— ± —	— ± —	— ± —	— ± —
ResNet-18/TinyImageNet-200	Standard	— ± —	— ± —	— ± —	— ± —
ResNet-18/TinyImageNet-200	Adversarial	— ± —	— ± —	— ± —	— ± —
ResNet-18/CUB-200-2011	Standard	— ± —	— ± —	— ± —	— ± —
ResNet-18/CUB-200-2011	Adversarial	— ± —	— ± —	— ± —	— ± —
ViT-B/16/CUB-200-2011	Standard	— ± —	— ± —	— ± —	— ± —
ViT-B/16/CUB-200-2011	Adversarial	— ± —	— ± —	— ± —	— ± —

Table 1. Summary statistics (mean ± 95% CI) for accuracy, PGD accuracy, S, and integral, by pairing and training regime, averaged across replicate runs.

Table 2

Pairing	Regime	Acc. Best	Acc. Last	PGD Best	PGD Last	S Best	S Last
ResNet-18/CIFAR-10	Standard	—	—	—	—	—	—
ResNet-18/CIFAR-10	Adversarial	—	—	—	—	—	—
VGG-16/CIFAR-10	Standard	—	—	—	—	—	—
VGG-16/CIFAR-10	Adversarial	—	—	—	—	—	—
ResNet-18/CIFAR-100	Standard	—	—	—	—	—	—
ResNet-18/CIFAR-100	Adversarial	—	—	—	—	—	—
WideResNet-28-10/CIFAR-100	Standard	—	—	—	—	—	—
WideResNet-28-10/CIFAR-100	Adversarial	—	—	—	—	—	—
ResNet-18/TinyImageNet-200	Standard	—	—	—	—	—	—
ResNet-18/TinyImageNet-200	Adversarial	—	—	—	—	—	—
ResNet-18/CUB-200-2011	Standard	—	—	—	—	—	—
ResNet-18/CUB-200-2011	Adversarial	—	—	—	—	—	—
ViT-B/16/CUB-200-2011	Standard	—	—	—	—	—	—
ViT-B/16/CUB-200-2011	Adversarial	—	—	—	—	—	—

Table 2. Best-checkpoint vs. last-checkpoint values, addressing potential drift/overfitting of the interpretability score over training.

Table 3

Pairing	Regime	Compute (h)	Runs (n)
ResNet-18/CIFAR-10	Standard	— ± —	2
ResNet-18/CIFAR-10	Adversarial	— ± —	2
VGG-16/CIFAR-10	Standard	— ± —	2
VGG-16/CIFAR-10	Adversarial	— ± —	2
ResNet-18/CIFAR-100	Standard	— ± —	2
ResNet-18/CIFAR-100	Adversarial	— ± —	2
WideResNet-28-10/CIFAR-100	Standard	— ± —	2
WideResNet-28-10/CIFAR-100	Adversarial	— ± —	2
ResNet-18/TinyImageNet-200	Standard	— ± —	2
ResNet-18/TinyImageNet-200	Adversarial	— ± —	2
ResNet-18/CUB-200-2011	Standard	— ± —	2
ResNet-18/CUB-200-2011	Adversarial	— ± —	2
ViT-B/16/CUB-200-2011	Standard	— ± —	2
ViT-B/16/CUB-200-2011	Adversarial	— ± —	2

Table 3. Wall-clock compute time per run, by pairing and regime.

Table 4

Hyperparameter value	Accuracy (%)	PGD Accuracy (%)	S	Integral	Compute (h)
Layer subset A	— ± —	— ± —	— ± —	— ± —	— ± —
Layer subset B	— ± —	— ± —	— ± —	— ± —	— ± —
Layer subset C	— ± —	— ± —	— ± —	— ± —	— ± —
Full rank	— ± —	— ± —	— ± —	— ± —	— ± —
Truncated rank	— ± —	— ± —	— ± —	— ± —	— ± —

Table 4. Ablation on the hyperparameter used to compute S (e.g., layer subset, rank truncation), showing its downstream effect on all tracked metrics.

Table 5

Pairing	Metric	Test statistic	p-value	Significant ($\alpha=0.05$)
ResNet-18/CIFAR-10	Accuracy	—	—	—
ResNet-18/CIFAR-10	PGD Accuracy	—	—	—
ResNet-18/CIFAR-10	S	—	—	—
ResNet-18/CIFAR-10	Integral	—	—	—
VGG-16/CIFAR-10	Accuracy	—	—	—
VGG-16/CIFAR-10	PGD Accuracy	—	—	—
VGG-16/CIFAR-10	S	—	—	—
VGG-16/CIFAR-10	Integral	—	—	—
ResNet-18/CIFAR-100	Accuracy	—	—	—
ResNet-18/CIFAR-100	PGD Accuracy	—	—	—
ResNet-18/CIFAR-100	S	—	—	—
ResNet-18/CIFAR-100	Integral	—	—	—
WideResNet-28-10/CIFAR-100	Accuracy	—	—	—
WideResNet-28-10/CIFAR-100	PGD Accuracy	—	—	—
WideResNet-28-10/CIFAR-100	S	—	—	—
WideResNet-28-10/CIFAR-100	Integral	—	—	—
ResNet-18/TinyImageNet-200	Accuracy	—	—	—
ResNet-18/TinyImageNet-200	PGD Accuracy	—	—	—
ResNet-18/TinyImageNet-200	S	—	—	—
ResNet-18/TinyImageNet-200	Integral	—	—	—
ResNet-18/CUB-200-2011	Accuracy	—	—	—
ResNet-18/CUB-200-2011	PGD Accuracy	—	—	—
ResNet-18/CUB-200-2011	S	—	—	—
ResNet-18/CUB-200-2011	Integral	—	—	—
ViT-B/16/CUB-200-2011	Accuracy	—	—	—
ViT-B/16/CUB-200-2011	PGD Accuracy	—	—	—
ViT-B/16/CUB-200-2011	S	—	—	—
ViT-B/16/CUB-200-2011	Integral	—	—	—

Table 5. Paired significance test (Wilcoxon signed-rank / paired t-test) per pairing and metric, testing whether adversarial training changes the metric relative to standard training.

Figure 1

Figure 1. Training curves across metrics and pairings (placeholder synthetic data)

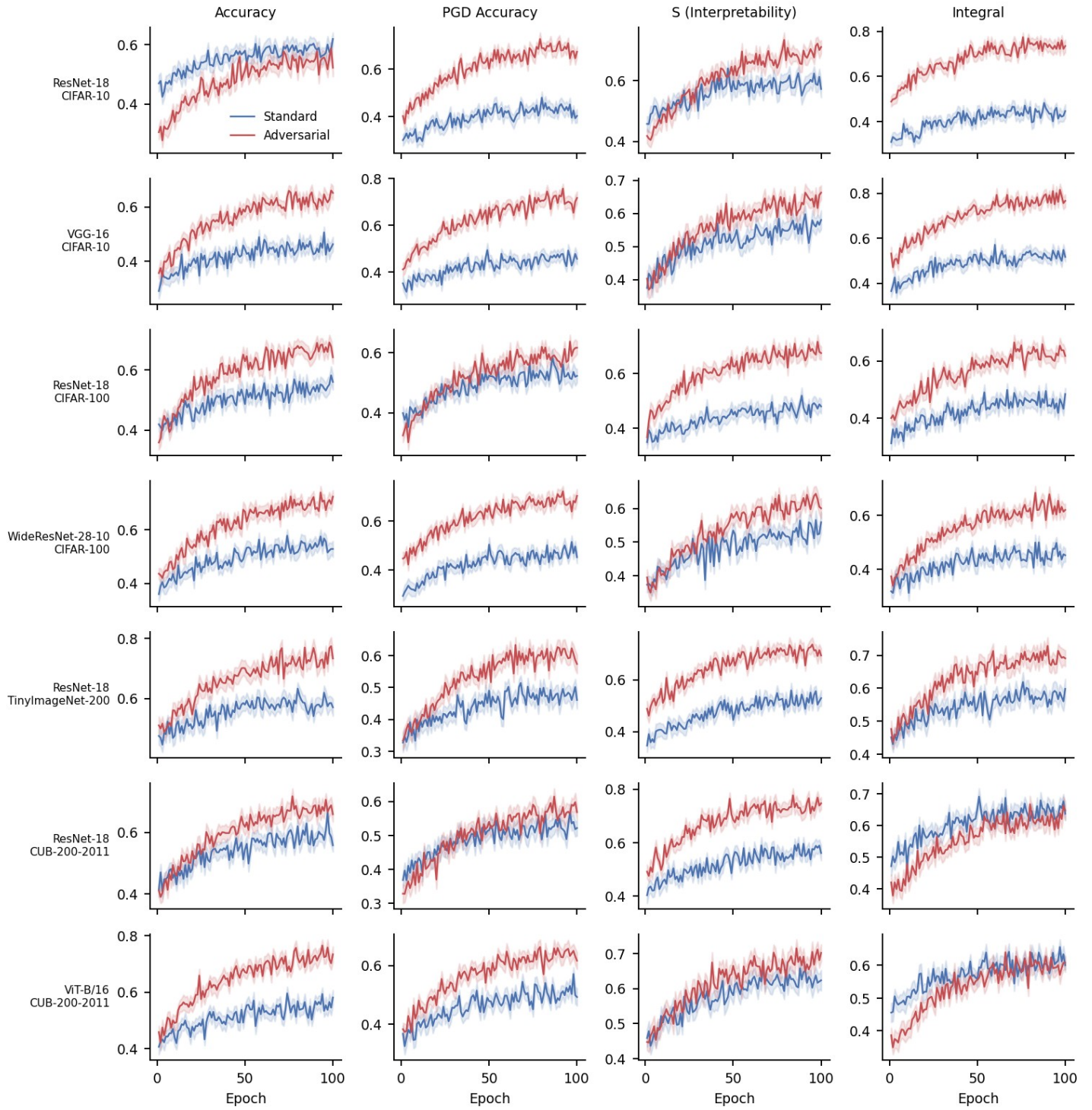


Figure 1. Training curves across metrics (accuracy, PGD accuracy, S, integral) and pairings, standard vs. adversarial. Shaded bands show ± 1 s.d. across replicate runs. Placeholder synthetic data.

Figure 2

Figure 2. Stability of S across replicate runs, standard (top) vs. adversarial (bottom) (placeholder synthetic data)

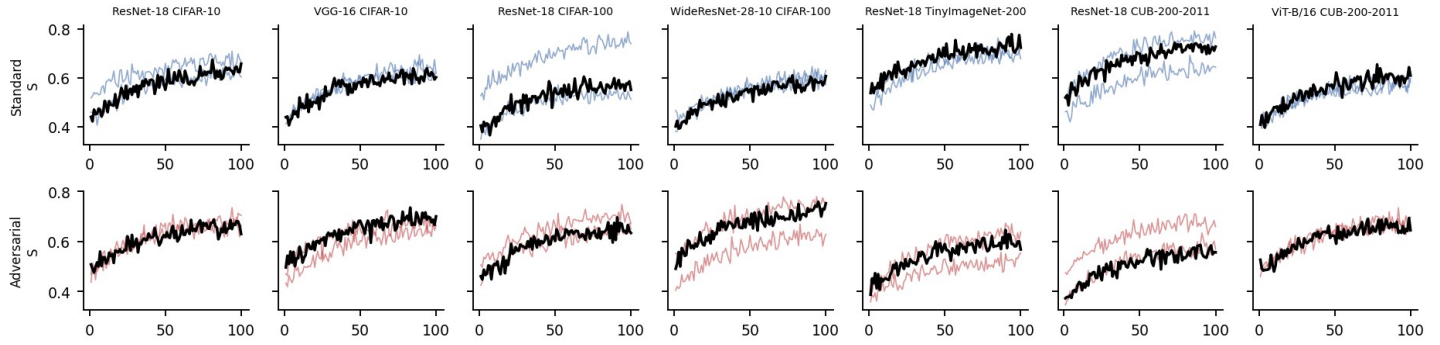


Figure 2. Stability of S across replicate runs, standard (top row) vs. adversarial (bottom row), by pairing. Thin lines: individual runs; thick line: mean. Placeholder synthetic data.

Figure 3

Figure 3. Accuracy $\times$ robustness $\times$ interpretability trade-offs (placeholder synthetic data)

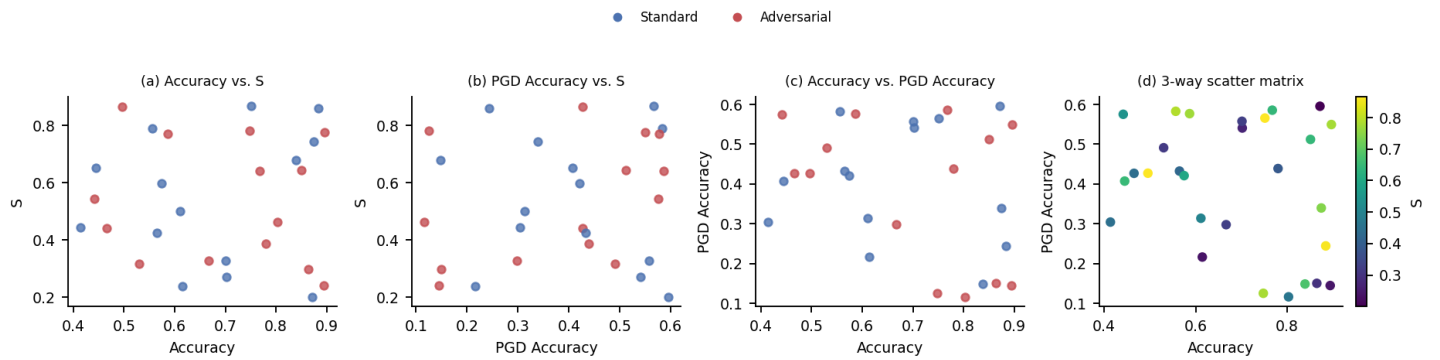


Figure 3. Accuracy $\times$ robustness $\times$ interpretability trade-offs. (a) Accuracy vs. S . (b) PGD accuracy vs. S . (c) Accuracy vs. PGD accuracy. (d) Joint scatter colored by S . Placeholder synthetic data.

Figure 4

Figure 4. Per-pairing distributions with raw observations overlaid (9.1) (placeholder synthetic data)

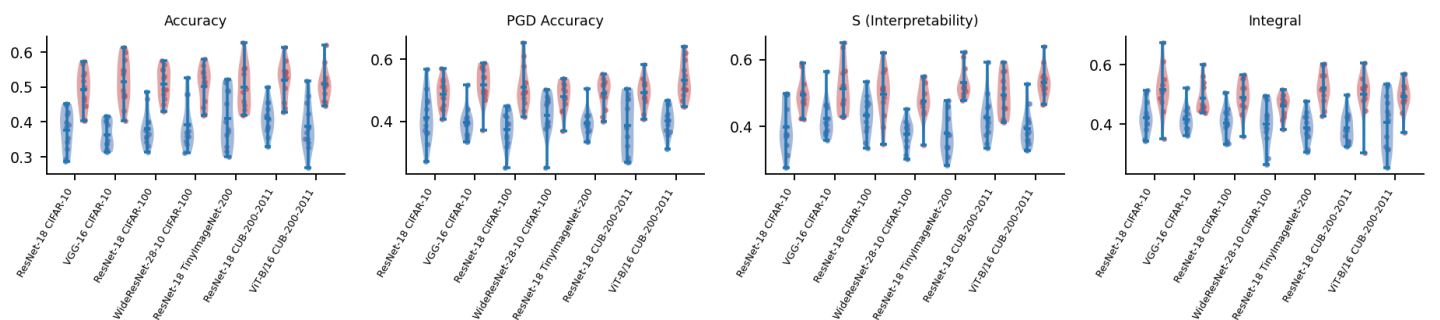


Figure 4 (9.1). Per-pairing distributions of accuracy, PGD accuracy, S , and integral, standard vs. adversarial, with raw observations overlaid on violin plots. Placeholder synthetic data.

Figure 5

Figure 5. Covariance eigenvalue spectra, standard vs. adversarial (9.2) (placeholder synthetic data)

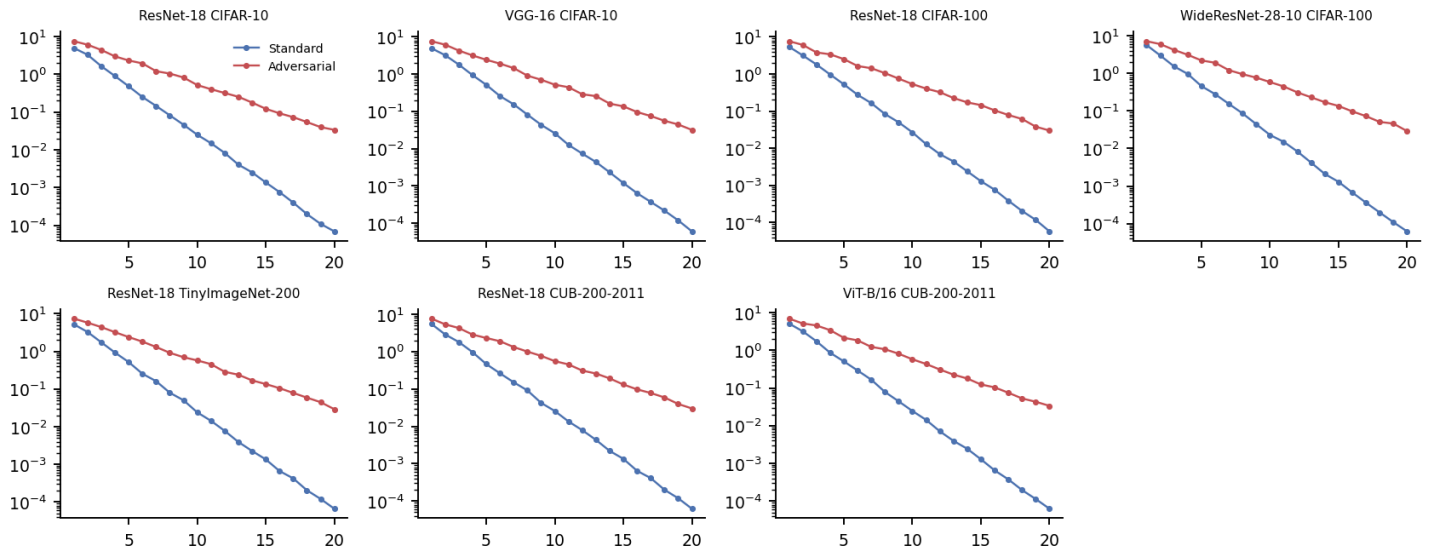


Figure 5 (9.2). Covariance eigenvalue spectra, standard vs. adversarial, by pairing — the raw material from which S is computed. Placeholder synthetic data.

Figure 6

Figure 6. Pairwise relationship scatter matrix, all 28 runs (9.5) (placeholder synthetic data)

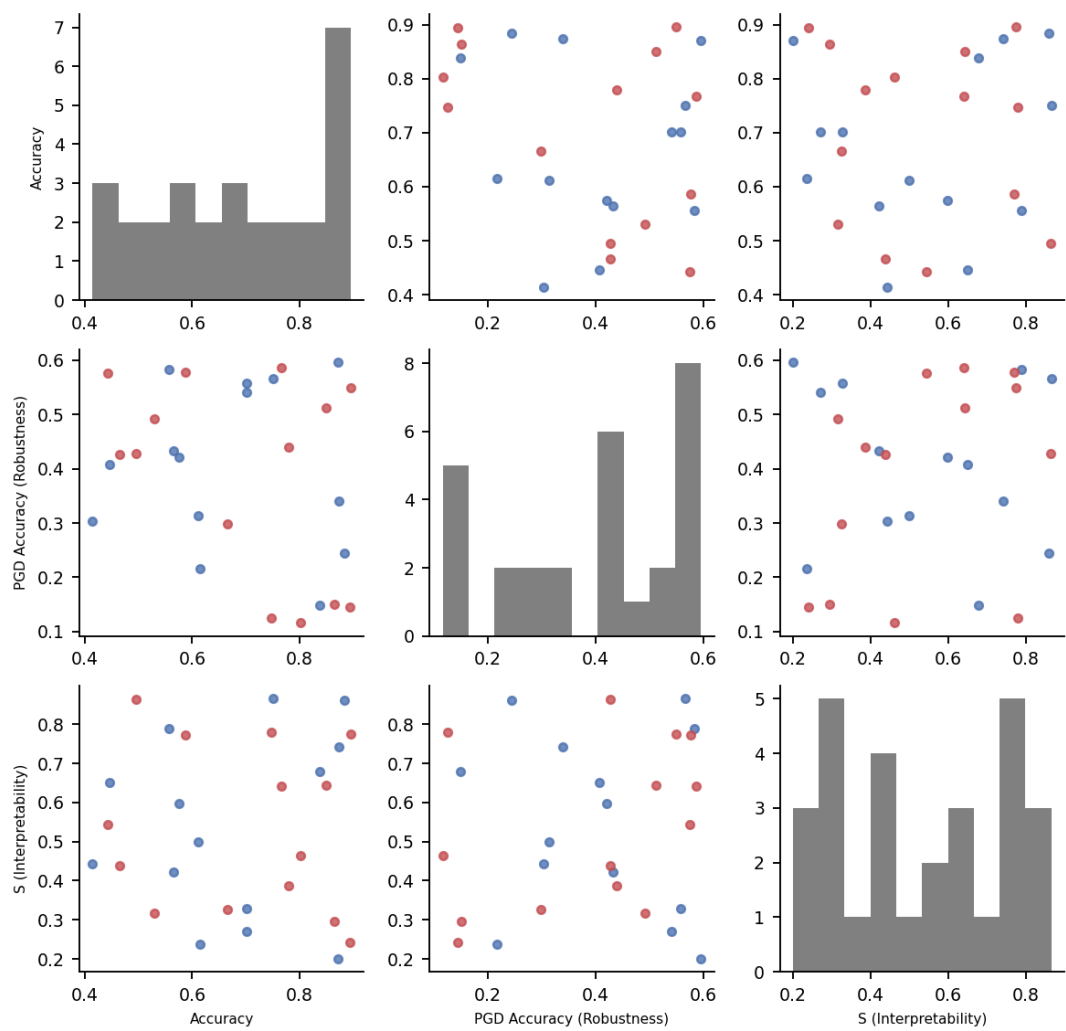


Figure 6 (9.5). Pairwise relationship scatter matrix — accuracy, PGD accuracy (robustness), and S (interpretability) — all individual runs plotted, colored by regime. Diagonal shows marginal histograms. Placeholder synthetic data.

Figure 7

Figure 7. Best (●) vs. last (x) checkpoint, per pairing/regime (9.6) (placeholder synthetic data)

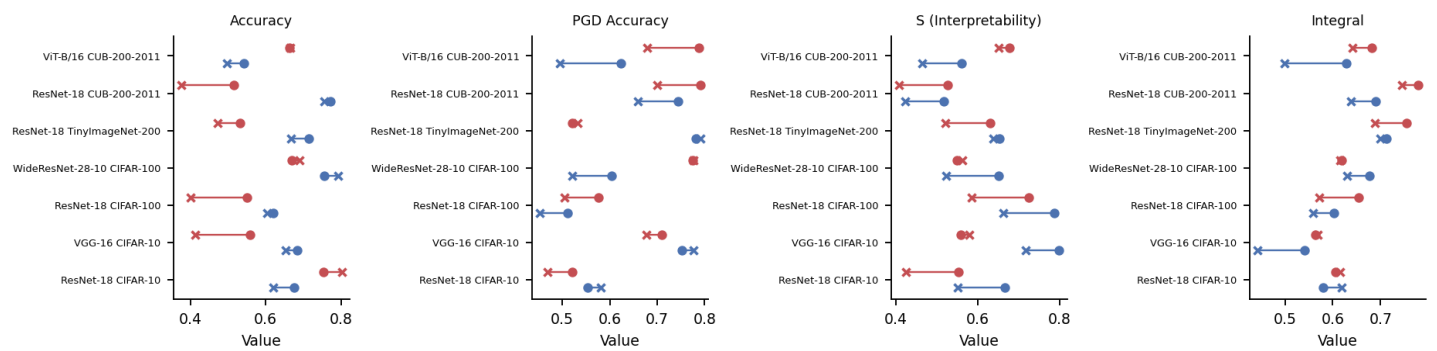


Figure 7 (9.6). Best (●) vs. last (x) checkpoint per pairing and regime, across metrics — drift/overfitting visible as connecting-line slope. Placeholder synthetic data.

Figure 8

Figure 8. Layer-wise score matrix: standard, adversarial, and difference (9.7) (placeholder synthetic data)

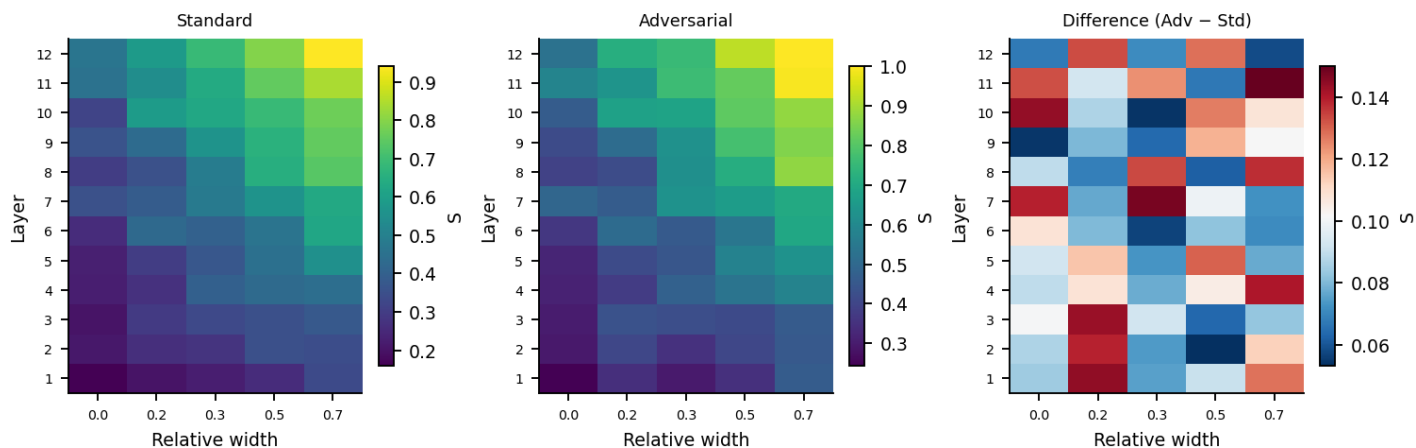


Figure 8 (9.7). Layer-wise S score matrix (layer $\times$ relative width): standard, adversarial, and difference (adversarial - standard). Placeholder synthetic data.

Figure 9

Figure 9. Layer $\times$ width $\times$ integral relationship (9.8) (placeholder synthetic data)

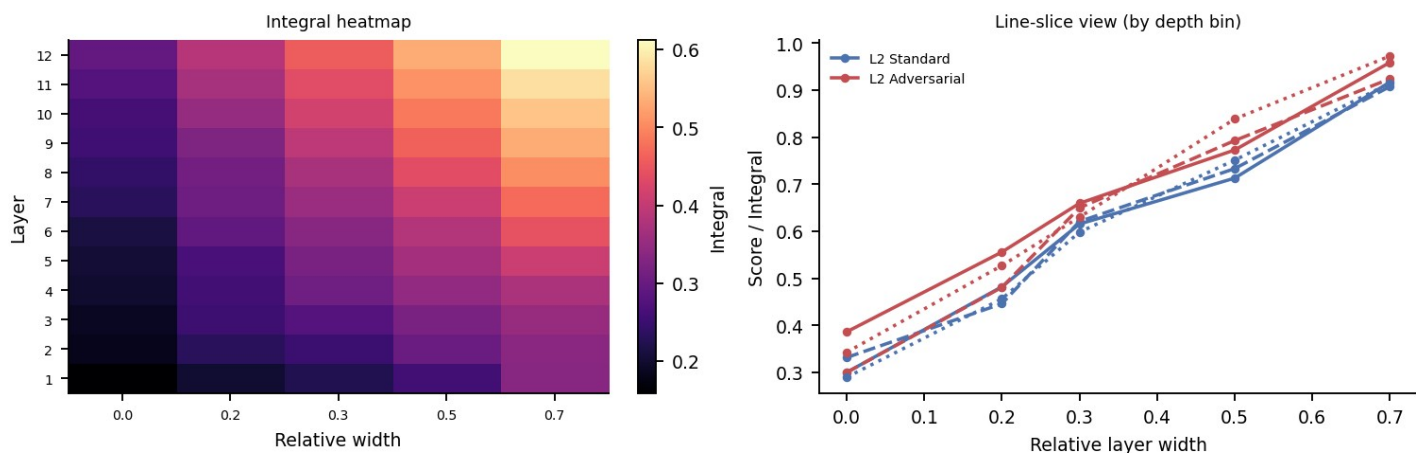


Figure 9 (9.8). Layer $\times$ width $\times$ integral relationship: heatmap of cumulative integral (left), and line-slice view by depth bin (right). Placeholder synthetic data.

Figure 10

Figure 10. Depth and width marginal profiles (9.9) (placeholder synthetic data)

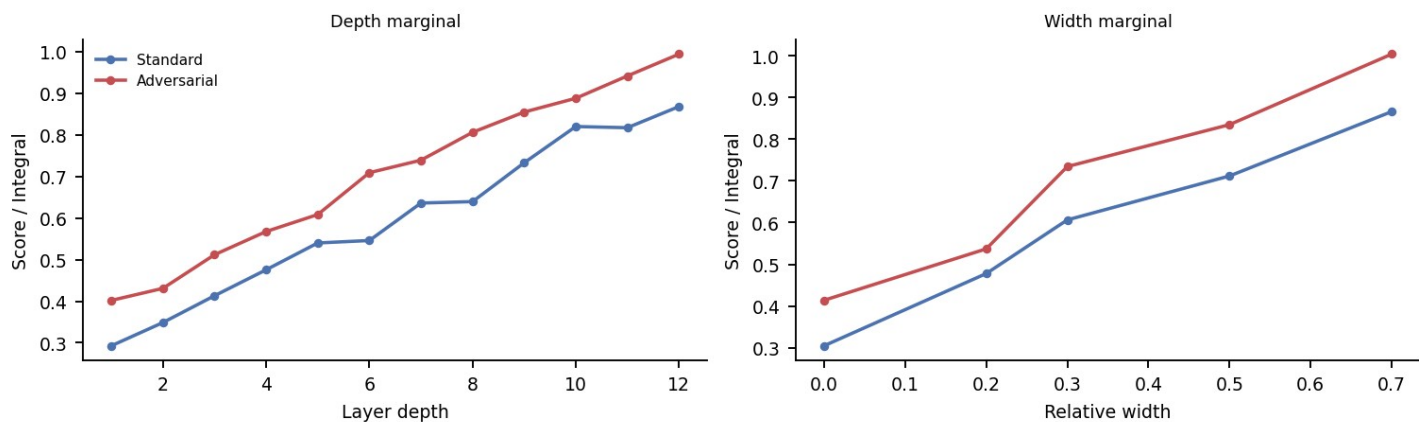


Figure 10 (9.9). Depth marginal (left) and width marginal (right) profiles of score/integral, standard vs. adversarial. Placeholder synthetic data.

Figure 11

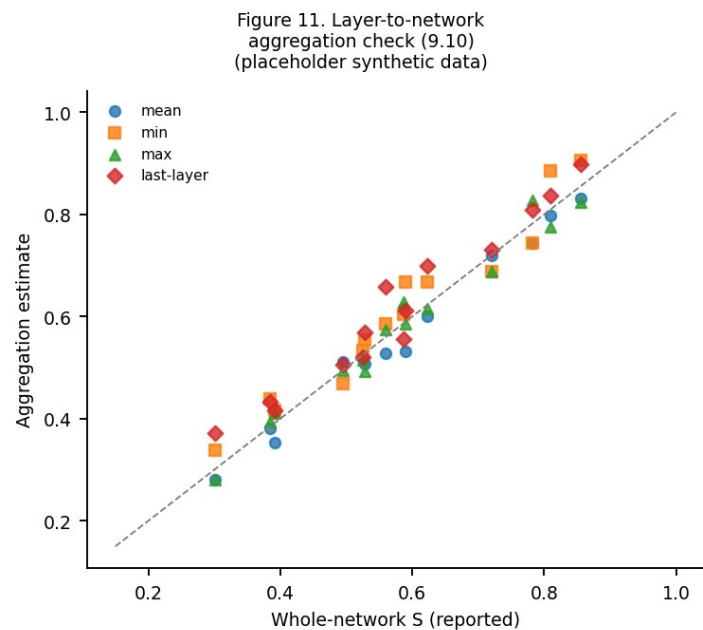


Figure 11 (9.10). Layer-to-network aggregation check: whole-network S (reported) vs. simple aggregations (mean, min, max, last-layer) of the per-layer matrix. Dashed line: identity. Placeholder synthetic data.

Figure 12

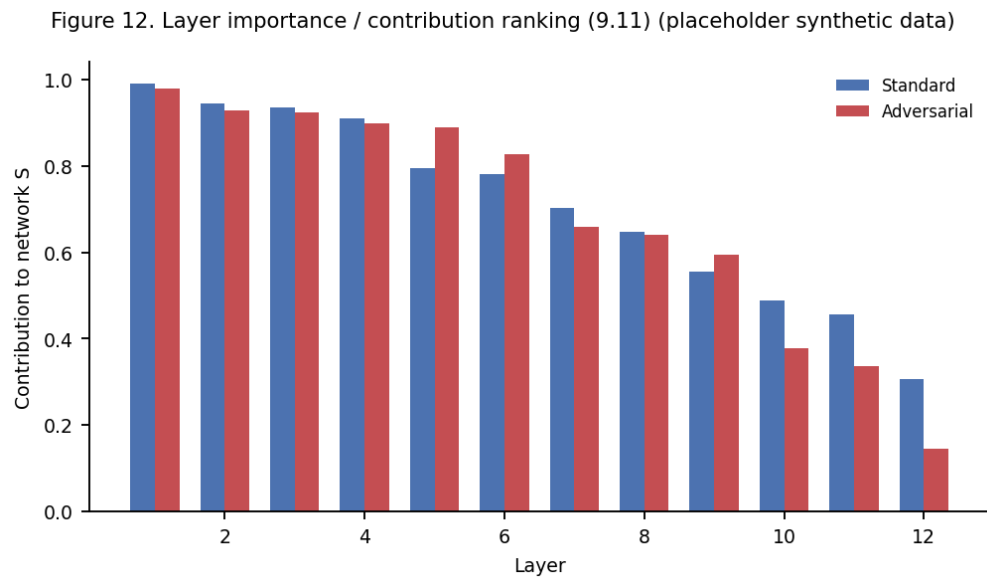


Figure 12 (9.11). Layer importance / contribution ranking to the aggregate network S, standard vs. adversarial. Placeholder synthetic data.

Figure 13

Figure 13. Cross-layer S correlation matrix (9.12) (placeholder synthetic data)

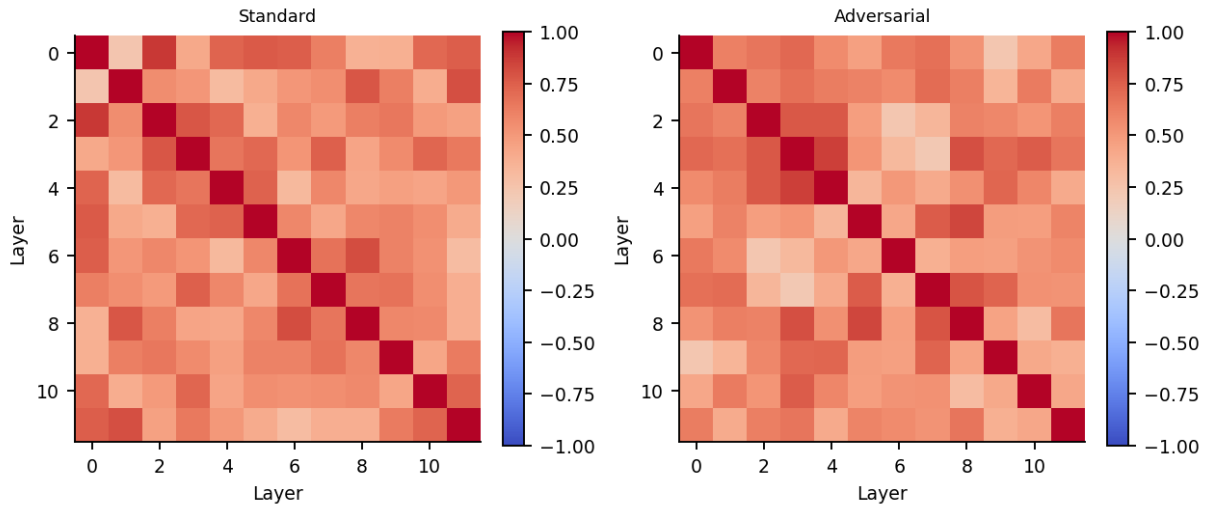


Figure 13 (9.12). Cross-layer correlation matrix of S values across runs/checkpoints, standard vs. adversarial. Placeholder synthetic data.

Figure 14

Figure 14. Layer-wise training trajectories of S (9.13) (placeholder synthetic data)

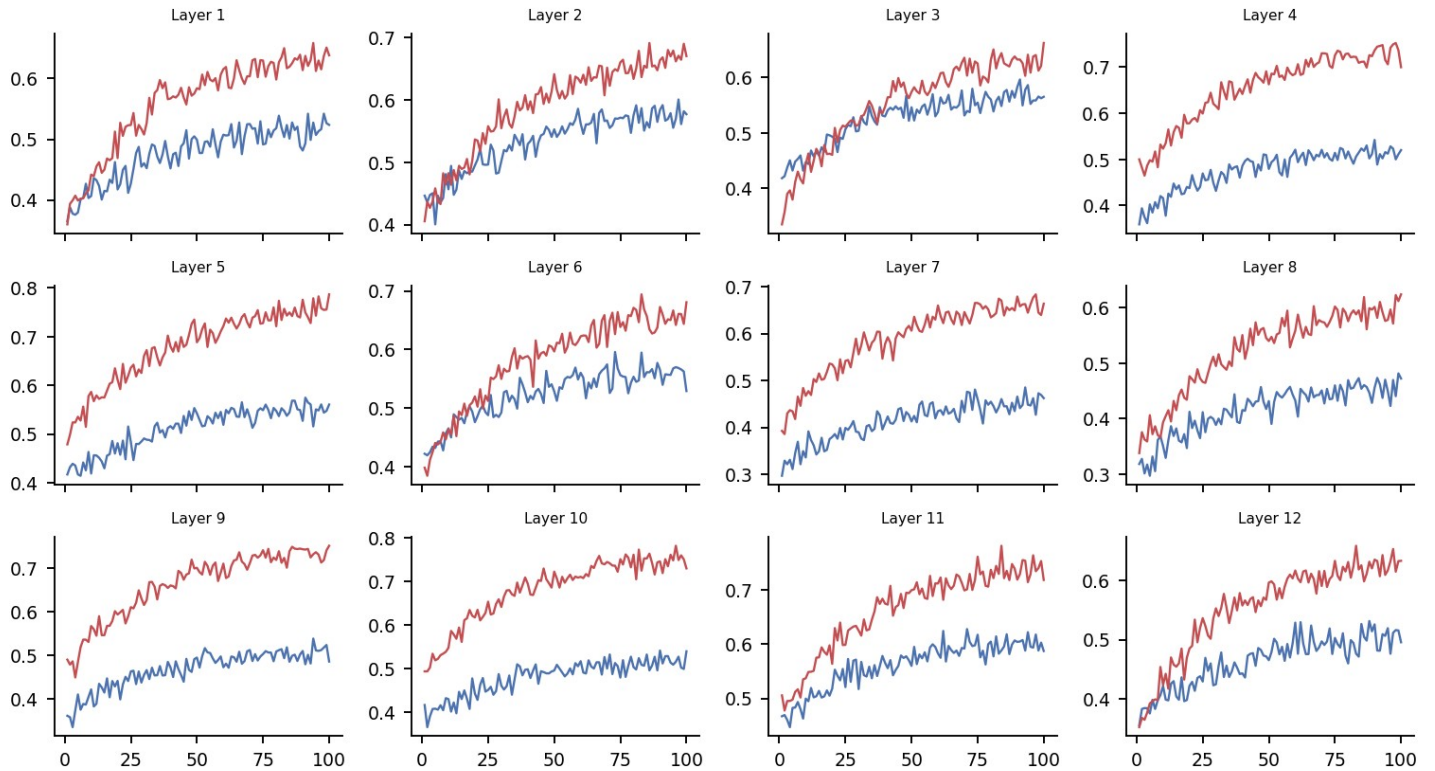


Figure 14 (9.13). Layer-wise training trajectories of S, one panel per layer, standard vs. adversarial. Placeholder synthetic data.

Figure 15

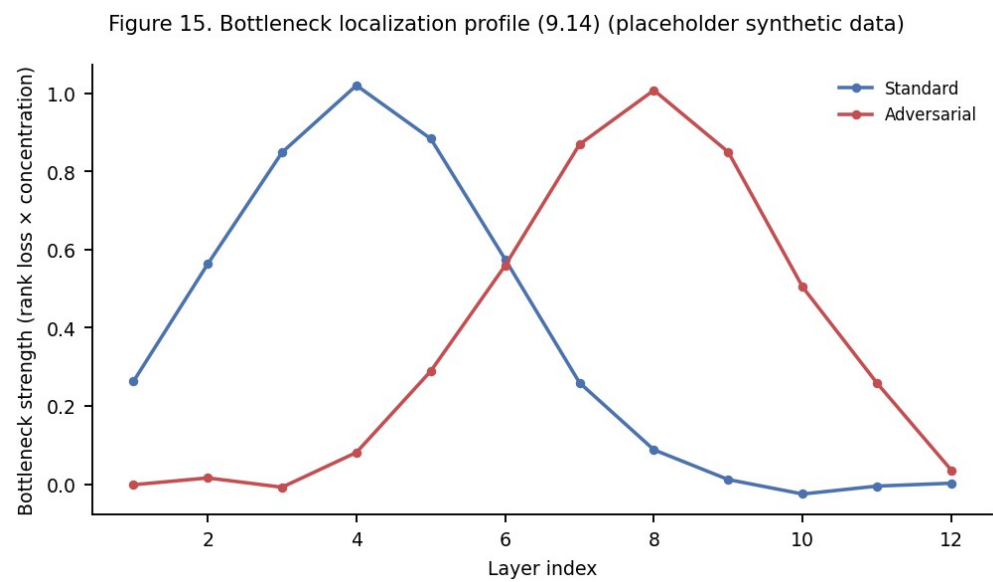


Figure 15 (9.14). Bottleneck localization profile across layer index (rank loss $\times$ concentration increase), standard vs. adversarial. Placeholder synthetic data.

Figure 16

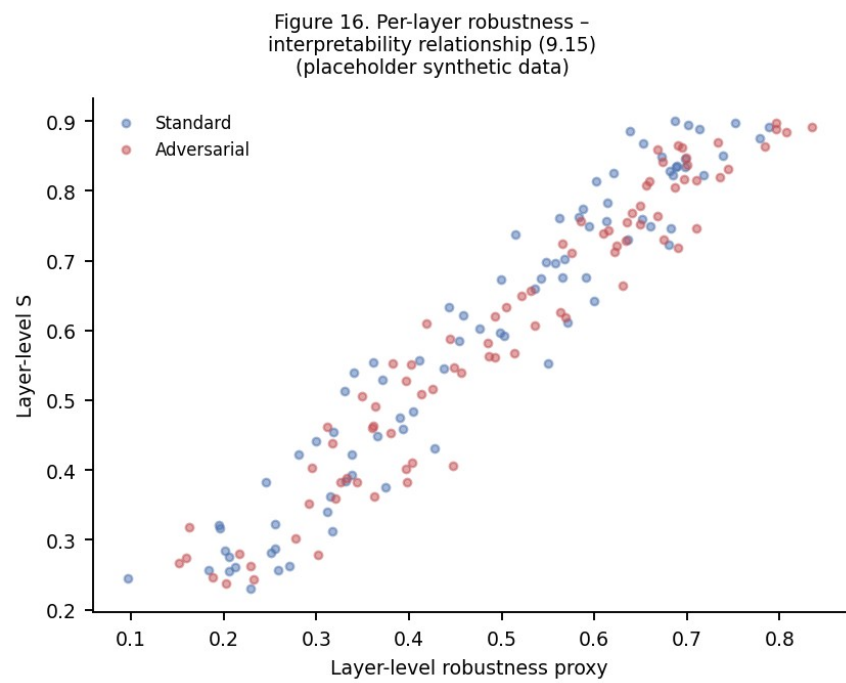


Figure 16 (9.15). Per-layer robustness–interpretability relationship: layer-level S vs. layer-level robustness proxy, standard vs. adversarial. Placeholder synthetic data.

Figure 17

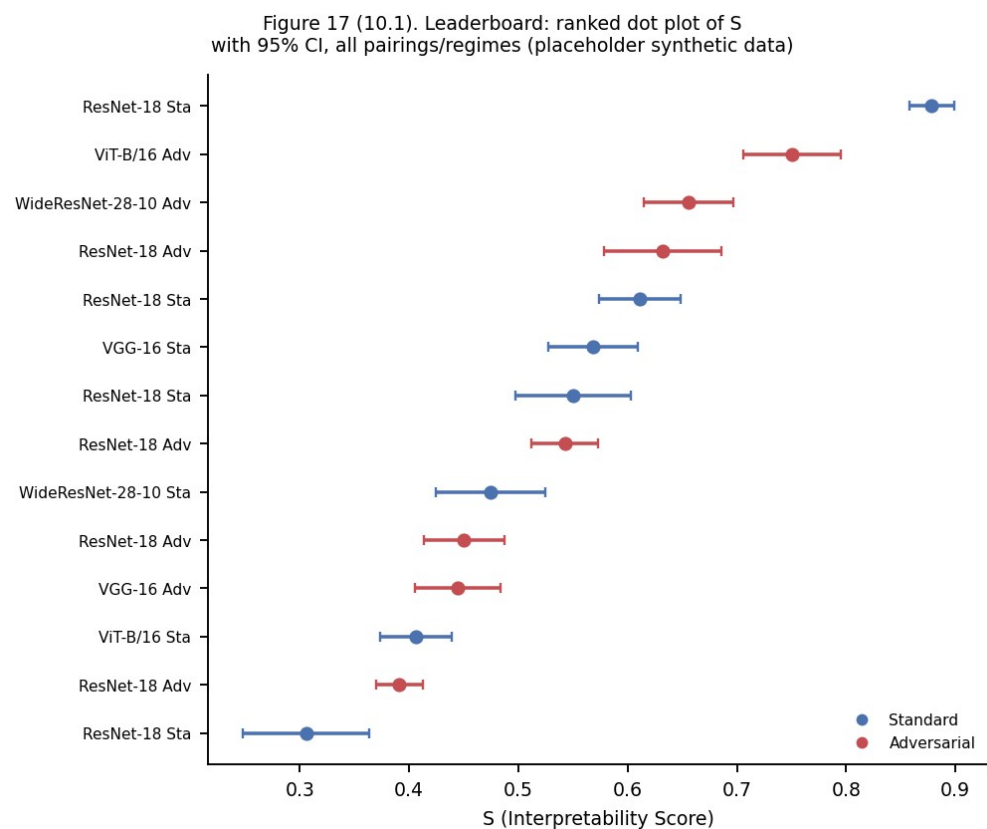


Figure 17 (10.1). Leaderboard: ranked dot plot of S with 95% CI, all pairings and regimes, sorted from highest to lowest. Placeholder synthetic data.

Figure 18

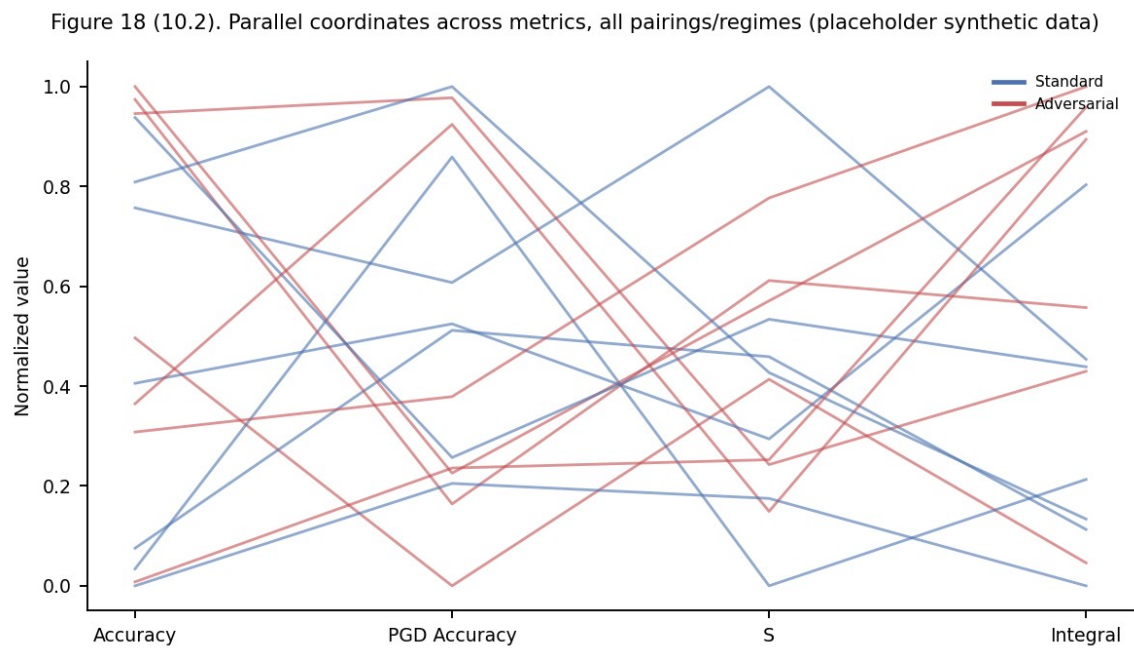


Figure 18 (10.2). Parallel coordinates across accuracy, PGD accuracy, S, and integral, one line per pairing/regime (normalized per axis). Placeholder synthetic data.

Figure 19

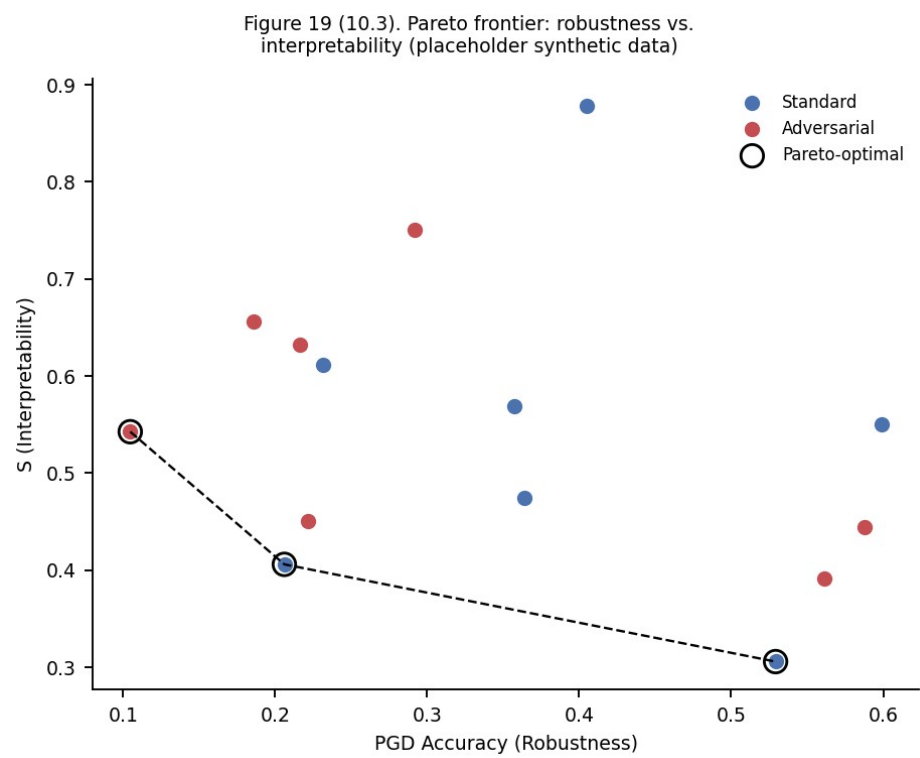


Figure 19 (10.3). Pareto frontier of robustness (PGD accuracy) vs. interpretability (S); open circles mark Pareto-optimal pairings/regimes. Placeholder synthetic data.

Figure 20

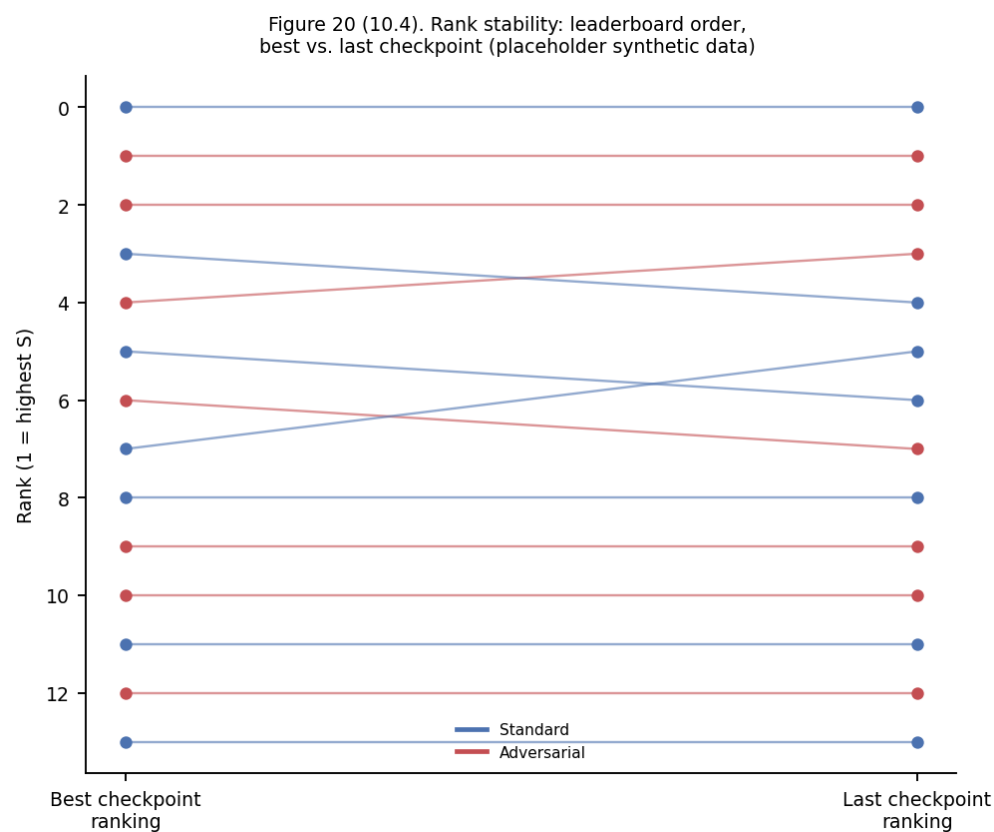


Figure 20 (10.4). Rank stability of the S leaderboard: ranking from best-checkpoint S vs. ranking from last-checkpoint S, per pairing/regime. Placeholder synthetic data.